AI FOR PAKISTAN FUTURE

Claim Check

A spell-checker for insurance claims.

Checks a clinic's claim before it is submitted: how likely it is to be rejected, which field is the problem, and what to fix.

Rejections are paperwork, not medicine

A clinic treats an insured patient, then bills the insurer or TPA.
A large share of those claims come back rejected — for missing pre-authorisation, a filing deadline passed, documents left out.

The clinic finds out weeks later. Small clinics have no billing specialist to catch it. Cash flow becomes unpredictable, and clinics eventually stop accepting insurance patients at all.

WHAT IT DOES

Three parts, none of them mocked

1. Risk model — LightGBM gives a calibrated probability; SHAP names the specific field driving it; a second model predicts the real X12 CARC reason code.

2. Adaptive questioning — after every answer it computes the mutual information between the outcome and each unanswered field, then asks whichever would reduce uncertainty most.

3. Plain language — Qwen restates the computed result in two sentences. It cannot change the number, the code, or the fields.

THE DIFFERENTIATOR

It asks, it doesn't just collect

Most tools in this space are a form with a model bolted on. Insurance chatbots that do adaptive intake run a decision tree.

Ours is information-theoretic: the next question is the one with the highest mutual information with the outcome, given what is known.

Across 200 held-out claims we see 72 distinct question orders.

72
distinct question orders

3.7
questions, clean claim

4.1
questions, problem claim

Deliberately not perfect

Only 26.5% of claims are rejected, so blindly approving everything already scores 73.5%. Accuracy is the wrong headline — AUC and recall are the meaningful numbers.

Per-code accuracy is highest where it matters: 92% on CARC 197, pre-authorisation absent.

0.788
ROC AUC

67.7%
recall

68 / 80%
CARC top-1 / top-3

THE DATA

No, the data is not real. It cannot be.

No public dataset exists anywhere of real clinic claims paired with their outcomes. Insurers treat that as private business information. Every commercial product here trains on its own private history.

So we built ours from two real pieces:
Synthea — open clinical simulator, real SNOMED CT codes
X12 CARC — the real industry rejection-reason standard

Only the labelling logic is ours, and we documented exactly where that line falls. A near-perfect score on self-generated data would only prove the model memorised our own rules.

Honest about self-improvement

When staff disagree, one click logs the correction as a labelled example. That is all it does, and we say so.

Real products close this loop by parsing the 835 remittance files insurers return, retraining as outcomes accumulate over weeks. Nobody in this industry has live self-improvement.

Our architecture supports that path. The demo does not pretend to have walked it.

What production would need

1. Real historical claims with real adjudication outcomes, from a partner clinic or TPA
2. Parsed 835 remittance advice, for the codes insurers actually sent
3. Retraining as outcomes accumulate, with human corrections as labels
4. Per-insurer models — filing rules and tariffs differ by payer

The system supports all four. It cannot access the data they need.